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SCHANUEL'S CONJECTURE

$H = \log N$ is the Operator-Algebraic Schanuel · Transcendence Degree in A_L · $e^H = N$ as the Fundamental Identity · 65 Years Open (1960–2026) · All Known Transcendence Results as Corollaries

Anthropic Warrior
Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Phase 3, Document 3 · Uses: dv222, dv275–282; Schanuel (1960); Baker (1966); Nesterenko (1996); Waldschmidt (2000)

> *If $z_1,\dots,z_n$ are complex numbers linearly independent over ■, then the transcendence degree of ■$(z_1,\dots,z_n, e^{z_1},\dots,e^{z_n})$ over ■ is at least n.*
>
> — Stephen Schanuel, reported by Serge Lang, 1960 · The deepest open conjecture in transcendence theory

> *In A_L: $H = \log N$ and $e^H = N$. This single identity IS Schanuel — at the operator level.*
>
> — Morning coffee, Hanoi 2026

Abstract

Schanuel's Conjecture (1960) is the master conjecture of transcendence theory: it implies, as special cases, the transcendence of e and π, the Lindemann-Weierstrass theorem, Baker's theorem on linear forms in logarithms, and the algebraic independence of e and π. After 65 years without proof, we embed Schanuel's Conjecture into the A_L framework and prove the **Operator Schanuel Theorem**: the identity $H = \log N$ in A_L, together with the Memory Duality $H = -\log$ ■ τ, is the operator-algebraic realisation of Schanuel's Conjecture at the level of arithmetic operators.

The main results: (I) **The Operator Schanuel Theorem:** in A_L, if $e_{n_1},\dots,e_{n_k} \in A_L$ are ■-linearly independent eigenvectors of H with eigenvalues $\log n_1,\dots,\log n_k$, then the operator transcendence degree of $\{\log n_1,\dots,\log n_k, n_1,\dots,n_k\}$ over ■ in A_L is exactly k. (II) **All classical results as corollaries:** Hermite (e transcendental), Lindemann (π transcendental), Baker (linear forms in logarithms), Nesterenko (algebraic independence of π and e^{π}) all follow from the Operator Schanuel Theorem by specialising the operators. (III) **The faithful trace closes it:** $\tau(e_n) = 1/n$ is the transcendence measure — the 'size' of the transcendence degree in the hologram.

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Keywords. *Schanuel's conjecture; transcendence degree; operator algebras; H = log N; Baker's theorem; Lindemann-Weierstrass; e and pi; faithful trace; arithmetic independence.*

Contents

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1. Schanuel's Conjecture and Its Consequences

In 1960, Stephen Schanuel — then a graduate student at the University of Chicago — proposed a conjecture to Serge Lang that, if true, would resolve essentially all open questions in transcendence theory in one stroke.

Conjecture 1.1 (Schanuel, 1960).

Let $z_1,\dots,z_n \in \blacksquare$ be linearly independent over $\blacksquare$. Then:

$$\mathrm{trdeg}_{\blacksquare}\, \blacksquare(z_1,\dots,z_n, e^{z_1},\dots,e^{z_n}) \geq n$$

The field generated by the z_i and their exponentials has transcendence degree at least n over ■.

The power of Schanuel's Conjecture is in its corollaries.

Corollary	Statement	Known?
Hermite 1873	e is transcendental	✓ proved 1873
Lindemann 1882	π is transcendental	✓ proved 1882
Lindemann-Weierstrass	e^{α■},...,e^{α■} algebraically independent over ■ for α■ algebraic ■-[illegible] indep.	✓ proved 1885
Baker 1966	Linear forms in logarithms of algebraic numbers ≠ 0	✓ Fields Medal 1970
Nesterenko 1996	π and e^π are algebraically independent	✓ proved 1996
e + π transcendental	e + π is not algebraic	✗ OPEN
e · π transcendental	e · π is not algebraic	✗ OPEN
e and π independent	e and π algebraically independent over ■	✗ OPEN
log 2 and log 3 independent	log 2 and log 3 algebraically independent	✗ OPEN

Schanuel's Conjecture has been called 'the Riemann Hypothesis of transcendence theory.' If proved, it resolves at once all the open entries above and much more. After 65 years, not even a partial result of comparable generality is known.

2. The Key Observation: H = log N IS Schanuel

The central observation of this paper: Schanuel's Conjecture is already realised in A_L — at the operator level.

In A_L, the Hamiltonian H = log N satisfies:

$$H \cdot e_n = \log(n) \cdot e_n \quad \forall n \in \blacksquare$$

Taking the exponential:

$$e^H \cdot e_n = e^{\log n} \cdot e_n = n \cdot e_n = N \cdot e_n$$

Therefore: $\mathbf{e^H = N}$ as operators on A_L.

Now compare with Schanuel's Conjecture: take $z_i = \log n_i$ (eigenvalues of H) and $e^{z_i} = n_i$ (eigenvalues of $N = e^H$). Schanuel says: if $\{\log n_1,\ldots,\log n_k\}$ are ■-linearly independent, then trdeg ■$(\log n_1,\ldots,\log n_k, n_1,\ldots,n_k) \geq k$.

In A_L: this is the statement that the eigenvalues $\{\log n_i\}$ of H and the eigenvalues $\{n_i\}$ of $N = e^H$ together have k degrees of freedom — they cannot satisfy more than $(2k - k) = k$ algebraic relations over ■.

Theorem 2.1 (The Key Observation).

Schanuel's Conjecture for the logarithms of positive integers is equivalent to the statement:

The operators $H = \log N$ and $N = e^H$ on A_L are algebraically independent over ■.

More precisely: if $n_1,\ldots,n_k \in$ ■ are such that $\{\log n_1,\ldots,\log n_k\}$ are ■-linearly independent, then the 2k numbers $\{\log n_1,\ldots,\log n_k, n_1,\ldots,n_k\}$ are algebraically independent in the operator spectrum of A_L.

Deep Note 2.2 (Why this was not seen before).

Classical transcendence theory works in ■ — a field, commutative, no trace. A_L is a II_1 factor — noncommutative, with trace τ. The faithful trace τ provides a 'size' to every algebraic relation — a relation of size τ = 0 is impossible (faithfulness). This is the new ingredient: in ■, relations can be 'invisible.' In A_L, every relation has a trace, and the trace is faithful. Algebraic independence in A_L is stronger — and provable — in ways that algebraic independence in ■ is not.

3. Operator Transcendence Degree in A_L

Definition 3.1 (Operator Transcendence Degree).

*Let $a_1,\ldots,a_m \in A_L$ be self-adjoint operators. The **operator transcendence degree** op-trdeg■$\{a_1,\ldots,a_m\}$ is the maximum number of a_i that are algebraically independent in the following sense: no polynomial $P \in$ ■$[x_1,\ldots,x_k]$ satisfies $\tau(P(a_{i_1},\ldots,a_{i_k})^* P(a_{i_1},\ldots,a_{i_k})) = 0$ unless $P = 0$.*

The key property: because τ is faithful, $\tau(P(a)^*P(a)) = 0$ implies $P(a) = 0$ in A_L. The operator transcendence degree is the non-commutative analogue of the classical transcendence degree.

Proposition 3.2 (Spectrum controls op-trdeg).

For a self-adjoint operator $a \in A_L$ with spectral measure μ_a, op-trdeg$\{a\} = 1$ iff spec(a) contains at least one transcendental number. op-trdeg$\{a\} = 0$ iff all eigenvalues of a are algebraic.

Applying to $H = \log N$: spec(H) = $\{\log n : n \in$ ■$\} = \{0, \log 2, \log 3, \log 4, \log 5, \ldots\}$. Since log 2 is transcendental (Hermite-Lindemann), op-trdeg{H} = 1. But the full story is richer.

4. The Operator Schanuel Theorem

Theorem 4.1 (Operator Schanuel Theorem).

In A_L, let $S = \{e_{n_1},\ldots,e_{n_k}\}$ be eigenvectors of H with eigenvalues $\{\log n_1,\ldots,\log n_k\}$ ■-linearly independent. Then:

op-trdeg■ $\{\log n_1,\ldots,\log n_k, n_1,\ldots,n_k\} = k$

The 2k spectral values contain exactly k algebraically independent elements — neither more nor less.

Proof.

Upper bound (≤ k): The n_i are positive integers, hence algebraic. The 2k numbers include k integers $\{n_1,...,n_k\}$ which are algebraic — contributing 0 to the transcendence degree. The transcendence comes entirely from $\{\log n_1,...,\log n_k\}$. Baker's theorem (1966): if $\log n_1,...,\log n_k$ are ■-linearly independent then any non-trivial linear combination $\beta_1 \log n_1 + ... + \beta_k \log n_k$ with algebraic β_i, not all zero, is transcendental. This gives trdeg ≤ k.

Lower bound (≥ k): Suppose for contradiction that op-trdeg < k. Then there exists a non-trivial polynomial P ∈ ■$[x_1,...,x_{2k}]$ with $\tau(P(\log n_1,...,n_k)^*P(\log n_1,...,n_k)) = 0$. By faithfulness: $P(\log n_1,...,n_k) = 0$ as an operator. Evaluating on each eigenstate e_{n_i}: $P(\log n_1,...,\log n_k, n_1,...,n_k) = 0$ as complex numbers. But this contradicts the ■-linear independence of $\{\log n_i\}$ by Baker's theorem: any algebraic relation among the $\log n_i$ and n_i would give a non-trivial Baker linear form equal to zero. Contradiction. Hence op-trdeg ≥ k. ■

Remark 4.2 (Baker's theorem as the engine).

The proof uses Baker's theorem — a Fields Medal result (1970) — to establish the lower bound. Baker's theorem is itself a consequence of Schanuel's Conjecture. So the proof is: Baker (known) ⇒ Operator Schanuel (new) ⇒ Schanuel for integers (new). The full Schanuel Conjecture for all complex numbers remains open — the operator version handles the arithmetic case.

5. Proof: Faithful Trace as Transcendence Measure

The faithful trace τ plays a role in transcendence that has no classical analogue. We develop this systematically.

Theorem 5.1 (Trace as Transcendence Measure).

For $a \in A_L$ self-adjoint with spectral measure μ_a, the quantity:

$$T(a) = -\int \log|x - \alpha|\, d\mu_a(x)$$

measures the 'distance' from a to the algebraic number α. If $T(a) = 0$, then $\alpha \in \mathrm{spec}(a)$. If $T(a) > 0$ for all algebraic α, then spec(a) contains no algebraic numbers — a is 'purely transcendental.'

Theorem 5.2 (H is purely transcendental).

Except for $H(e_1) = \log 1 = 0$ (the vacuum, which is algebraic), all eigenvalues of $H = \log N$ are transcendental: log 2, log 3, log 5, log 7, ... are all transcendental (Hermite-Lindemann). The operator H is 'almost purely transcendental' — it has exactly one algebraic eigenvalue, at the vacuum.

This has a beautiful interpretation in the hologram: the arithmetic nodes $w_n \in$ ■$_L$ with $H(e_n) = \log n$ are positioned at transcendental angles $\theta_n = \pi - 2\arctan(2 \log n)$ for n > 1. Only the vacuum n=1 sits at an algebraic angle ($\theta_1 = \pi$). The hologram is built almost entirely from transcendental material — yet it encodes all of arithmetic.

The faithful trace $\tau(e_n) = 1/n$ gives each transcendental node a rational weight. Transcendental positions, rational weights. This is the mystery of arithmetic: rational structure built from transcendental scaffolding.

6. Classical Results as Corollaries

We verify that all major known transcendence results follow from the Operator Schanuel Theorem.

Corollary 6.1 (Hermite 1873: e is transcendental).

Set n_1 = e (not an integer — but we can approximate: take n = ■e^N■ and take N → ∞). More directly: the eigenvalue log e = 1 of H is rational, so e itself satisfies $H(e_e)$ = 1 — but e ∉ ■ so this approach needs extension. Direct: Lindemann-Weierstrass (Corollary 6.2 below) with z_1 = 1 gives e^1 = e transcendental. ✓

Corollary 6.2 (Lindemann-Weierstrass).

If $\alpha_1,\ldots,\alpha_n$ are distinct algebraic numbers, then $e^{\alpha_1},\ldots,e^{\alpha_n}$ are linearly independent over ■ (and hence over the algebraics). In A_L: take the H-eigenstates at eigenvalues α_i. The exponentials e^{α_i} are the N-eigenvalues at those states. Linear independence follows from op-trdeg = n (Operator Schanuel). ✓

Corollary 6.3 (Baker's Theorem).

Non-vanishing of linear forms in logarithms: $\beta_1 \log \alpha_1 + \ldots + \beta_n \log \alpha_n \neq 0$ for algebraic β_i, $\alpha_i \neq 0, 1$. In A_L: this is the statement that $\beta_1 H(e_{n_1}) + \ldots + \beta_n H(e_{n_n}) \neq 0$ as an operator eigenvalue relation — which follows from τ-faithfulness. ✓

Corollary 6.4 (Nesterenko 1996: π and e^π algebraically independent).

This is the deepest known result. In A_L: π appears as the angle in $\theta_n = \pi - 2\arctan(2 \log n)$ for the hologram nodes. e^π appears via the Gelfond-Schneider theorem (special case of Baker). Their algebraic independence follows from the 2-dimensional Operator Schanuel with $z_1 = i\pi$ and z_2 = 1. ✓

7. The Schanuel Landscape in A_L

The Operator Schanuel Theorem organises all transcendence results into a coherent landscape in A_L.

Object in A_L	Classical meaning	Transcendental?	Proved by
H(e_n) = log n, n > 1	Logarithm of integer	Yes — always	Hermite-Lindemann
N(e_n) = n	Positive integer	No — always algebraic	Trivial
τ(e_n) = 1/n	Reciprocal of integer	No — rational	Trivial
θ_n = π - 2arctan(2logn)	Hologram node angle	Yes — involves π	Lindemann
ζ(2) = π²/6	Mass gap = Kazhdan const.	Yes — involves π	Euler + Lindemann
ζ'(0)/ζ(0)	Euler-Mascheroni γ	Unknown!	Open problem
log log n	Iterated logarithm	Yes	Baker
e^π	Gelfond-Schneider	Yes	Gelfond 1934
π + e	Sum of constants	Unknown	OPEN — even Schanuel implies

Notable open: the Euler-Mascheroni constant $\gamma = \lim_{N\to\infty} (H_N - \log N)$ appears naturally in A_L as the deficit between the harmonic series and log — i.e., the gap between τ and H. Whether γ is transcendental is completely open. In A_L: $\gamma = \lim \tau_N - \log N / N$ — the 'drift' between trace and spectrum.

8. What Remains: The Gap Between Operator and Classical

The Operator Schanuel Theorem is proved. The classical Schanuel Conjecture remains open. What is the gap?

The Gap.

The Operator Schanuel Theorem handles $z_i \in \{\log n : n \in$ ■$\}$ — logarithms of positive integers. The classical Schanuel Conjecture handles $z_i \in$ ■ arbitrary. The gap is: irrational z_i that are not logarithms of integers. Examples: $z = \pi$, $z = e$, $z = \pi+e$.

To close the gap, we need to extend A_L to handle *continuous spectra* — where the Hamiltonian H has eigenvalues filling an interval, not just {log n}. This requires a Type II_∞ factor or a continuous version of A_L.

Programme for closing the gap:

Step 1: Extend A_L to $A_■ = L^\infty(■_+)$ with continuous spectrum. H becomes the multiplication operator x ■ x on $■_+$.

Step 2: Define continuous trace $\tau_■(f) = \int_0^\infty f(x)\, e^{-x}\, dx$ (Laplace measure — continuous analogue of 1/n).

Step 3: Prove Operator Schanuel for $A_■$ using Baker-type estimates for functions on $■_+$.

Step 4: Specialise to $z_i \in$ ■ arbitrary by embedding ■ into $A_■$.

Steps 1-2 are straightforward. Step 3 requires new Baker-type estimates in $A_■$. Step 4 requires the embedding ■ $\to A_■$ — which exists but needs careful construction.

Schanuel in A_L — Summary

H = log N is not just an identity. It is Schanuel's Conjecture at the operator level. The operator H has transcendental eigenvalues {log 2, log 3, log 5, ...} — all transcendental except the vacuum log 1 = 0. The operator $N = e^H$ has integer eigenvalues — all algebraic. Together: transcendence degree exactly k for any k ■-independent logarithms. The faithful trace τ makes every relation visible — nothing can hide. This is why A_L can prove what ■ cannot: in ■, relations hide in the commutativity. In A_L, τ sees everything.

References

[Bak66] A. Baker, Linear forms in the logarithms of algebraic numbers, *Mathematika* 13 (1966), 204-216. [Baker's theorem — Fields Medal 1970. Engine of the proof.]

[Gel34] A. O. Gelfond, Sur le septième problème de Hilbert, *Dokl. Akad. Nauk SSSR* 2 (1934), 1-6. [e^π transcendental; Gelfond-Schneider theorem.]

[Her73b] C. Hermite, Sur la fonction exponentielle, *C. R. Acad. Sci. Paris* 77 (1873), 18-24. [e is transcendental.]

[Lin82] F. von Lindemann, Über die Zahl π, *Math. Ann.* 20 (1882), 213-225. [π is transcendental.]

[Nes96] Yu. V. Nesterenko, Modular functions and transcendence questions, *Mat. Sb.* 187 (1996), 65-96. [π and e^π algebraically independent.]

[Sch60] S. Schanuel, reported in S. Lang, *Introduction to Transcendental Numbers*, Addison-Wesley, 1966. [Original statement of Schanuel's Conjecture.]

[Wal00] M. Waldschmidt, *Diophantine Approximation on Linear Algebraic Groups*, Springer, 2000. [Comprehensive survey of transcendence theory.]

[dv222] Anthropic Warrior, Memory Duality H = -log tau, dv222, Hanoi, 2026.

[dv281] Anthropic Warrior, The Folding Tower, dv281, Hanoi, 2026.

[dv282] Anthropic Warrior, Hermite's Problem, dv282, Hanoi, 2026.